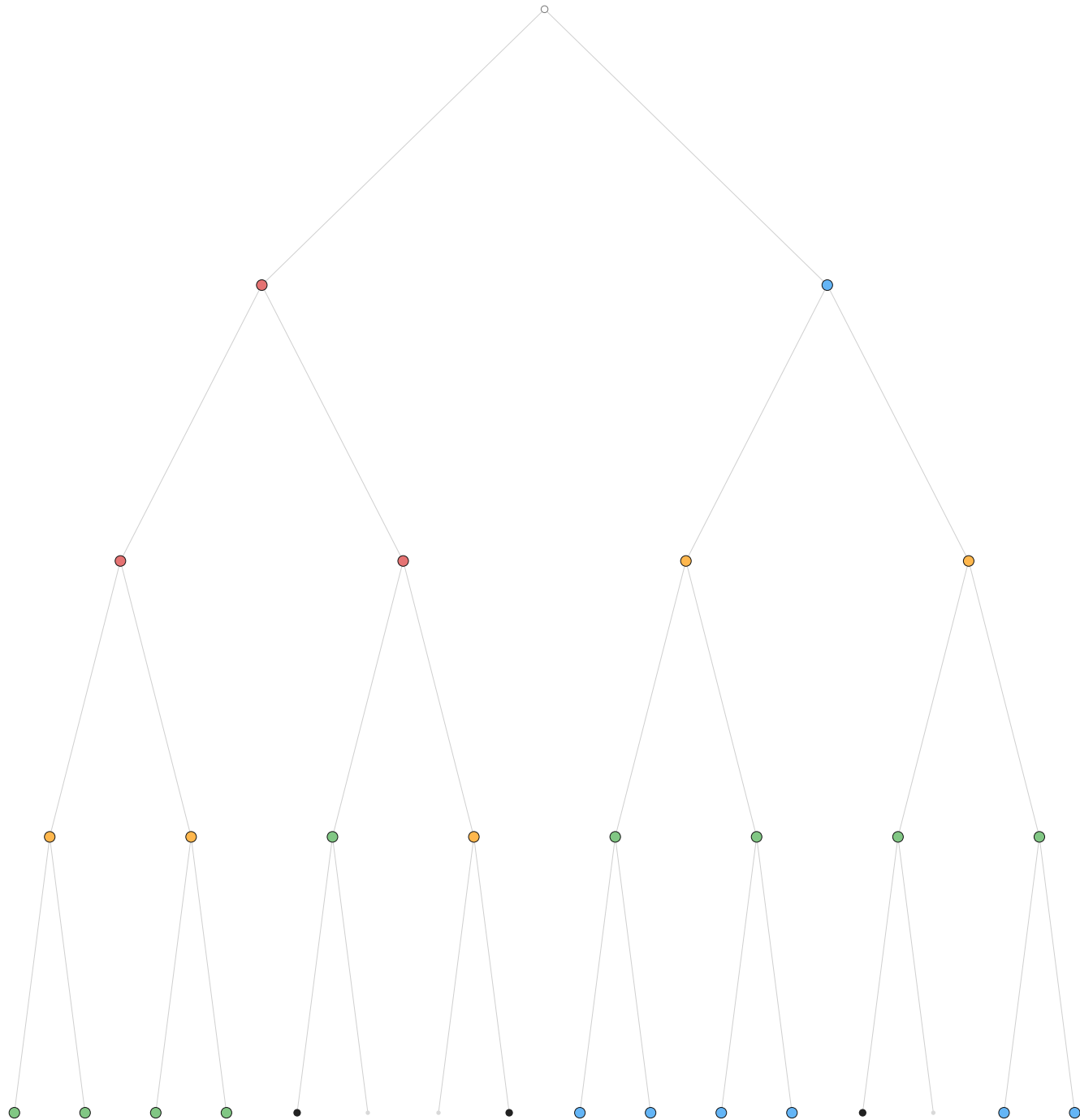


Complete Merkle tree (height 4) with stored proof-bearing nodes highlighted

- default empty node
- real but not stored
- occupied leaf
- stored node C1
- stored node C2
- stored node C3
- stored node C4



Example settings
height = 4
total leaves = 16
occupied leaves = 13
bottom-layer slots = [0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 14, 15]
sparsity = 18.8%

Coloring result
colors = 4
stored proof nodes = 24
count loads = [3, 5, 9, 7]
weight loads = [13, 12, 12, 12]
count gap = 6
weight gap = 1

Sample batch-PIR views
leaf heap=16 slot=0: C1->5, C2->9, C3->17, C4->3
leaf heap=24 slot=8: C1->2, C2->7, C3->13, C4->25
leaf heap=31 slot=15: C1->2, C2->6, C3->14, C4->30